



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI MALAI MADHESWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

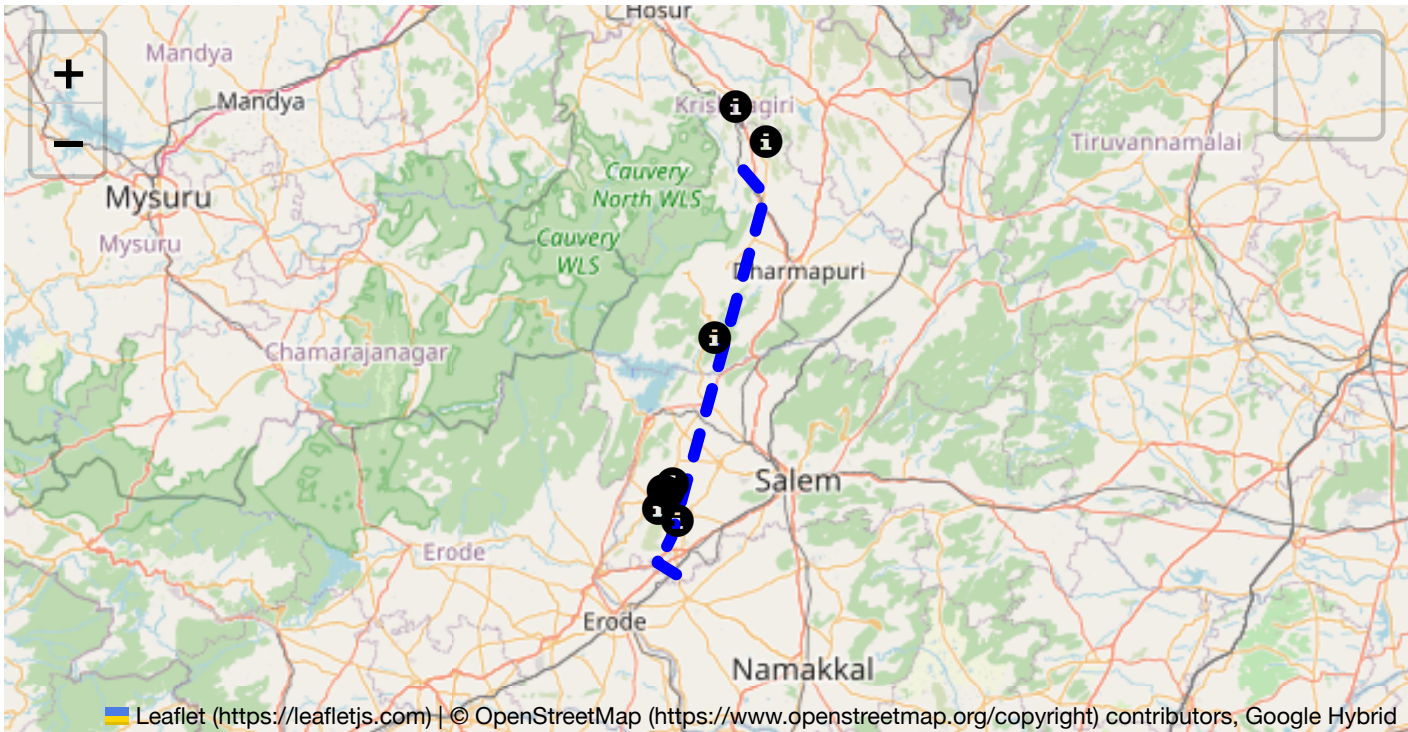
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

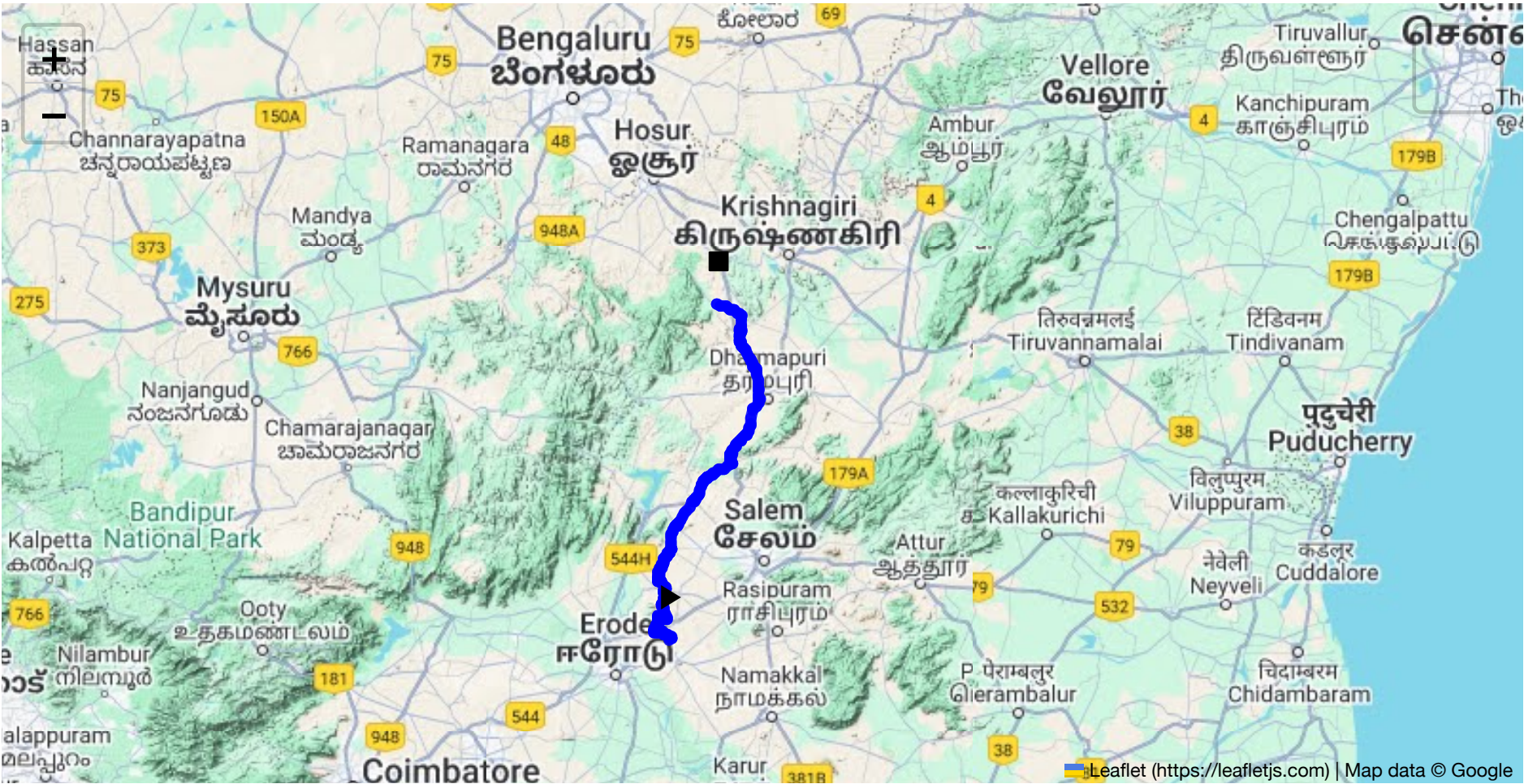


Route Summary:
Total Distance: 139.77 km
Estimated Duration: 3.2 hours

Adjusted Duration (Heavy Vehicle): 3.9 hours

Start: (11.4381, 77.8734)

End: (12.387217, 78.009354)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Chikkadornabettam, Tamil Nadu, covers approximately 140 kilometers. This journey passes through several smaller towns and villages, implying a mix of rural and semi-urban terrain. The path includes both state and national highways, with some segments potentially involving narrower local roads.

2. Typical Weather Conditions and Potential Hazards

This region experiences a tropical climate. The weather conditions vary significantly with seasons:

- **Summer (March to June):** Temperatures can soar, raising concerns over vehicle overheating.
- **Monsoon (June to September):** The region receives moderate to heavy rainfall, leading to slippery roads and potential flooding in low-lying areas.
- **Winter (October to February):** Milder weather conditions; fog can reduce visibility in the early mornings.

3. Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) see increased local traffic as people commute to work and back.
- **Congestion-Prone Areas:** Town centers, particularly around Kanakankadu and Idappadi, may experience congestion due to local market activities and narrow streets.

4. Road Quality and Infrastructure

- **State Highways:** Generally well-maintained but may have potholes, particularly after the monsoon.

- **Local Roads:** Quality varies; some segments might be poorly maintained, with uneven surfaces in rural areas.
- **Infrastructure:** Adequate signage in urban areas, but might be lacking in rural stretches; street lighting varies.

5. Alternative Routes for Emergencies

- **Via NH79:** Offers a more direct alternative if starting congestion occurs but diverts significantly.
- **Local Bypasses:** Use small link roads between major stops to avoid congested town centers, though these may not be documented thoroughly.

6. Local Regulations on Hazardous Material Transport

- Restriction during daytime in certain towns to minimize risk during peak pedestrian movement.
- Mandatory signage on vehicles carrying hazardous materials.
- Specific permits and notifications required for passage through urban areas.

7. Historical Incidents

- **Accidents:** Records indicate occasional vehicle mishaps due to road conditions or weather.
- **Hazardous Spills:** No major recorded incidents; however, local communities may lack preparedness for large-scale hazardous material spills.

8. Environmental Considerations

- Passes through agricultural areas; chemical spills could impact local water sources and crops.
- Proximity to certain wildlife areas; ensure minimal disturbance to local flora and fauna.

9. Communication Coverage

- **Urban Areas:** Good mobile coverage with occasional interruptions.
- **Rural Segments:** Potential dead zones, especially in more remote stretches like near forested sections.

10. Estimated Emergency Response Times

- **Urban Areas:** Quick response due to proximity to local police and fire stations (approximately 10-20 minutes).
- **Rural Areas:** Longer response times due to distance, potentially 30-45 minutes or more.

11. Overall Summary of Risk Assessment

This route is generally safe but poses some risks, especially in terms of road conditions, and weather-related hazards. Congestion in towns can delay transport, particularly during peak hours. It's vital to plan travel around local conditions and restrictions, maintaining open communication and considering alternative routes for emergencies. Familiarity with local climate, regulations, and infrastructure will enhance safety for truck drivers transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	High	11.45708, 77.82867	15 KM/Hr
9	Turn	High	11.46547, 77.82486	15 KM/Hr
10	Turn	Medium	11.47251, 77.83307	30 KM/Hr
11	Turn	High	11.47574, 77.83491	15 KM/Hr
12	Turn	High	11.47614, 77.83416	15 KM/Hr
13	Turn	High	11.48033, 77.83503	15 KM/Hr
14	Turn	High	11.48043, 77.83583	15 KM/Hr
15	Blind Spot	Blind Spot	11.50430, 77.83818	10 KM/Hr
16	Turn	Medium	11.50074, 77.84550	30 KM/Hr
17	Turn	High	11.49946, 77.84592	15 KM/Hr
18	Blind Spot	Blind Spot	11.49855, 77.86542	10 KM/Hr
19	Turn	Medium	11.50898, 77.86029	30 KM/Hr
20	Turn	Medium	11.52519, 77.85535	30 KM/Hr
21	Turn	Medium	11.55499, 77.85656	30 KM/Hr
22	Turn	Medium	11.55605, 77.85768	30 KM/Hr
23	Turn	Medium	11.56378, 77.85967	30 KM/Hr
24	Turn	High	11.57893, 77.85430	15 KM/Hr
25	Turn	Medium	11.57919, 77.85585	30 KM/Hr
26	Turn	Medium	11.58037, 77.85749	30 KM/Hr
27	Turn	Medium	11.60225, 77.84735	30 KM/Hr
28	Turn	Medium	11.62707, 77.84192	30 KM/Hr
29	Turn	Medium	11.63085, 77.84072	30 KM/Hr
30	Turn	Medium	11.64073, 77.84627	30 KM/Hr
31	Turn	Medium	11.64130, 77.84816	30 KM/Hr
32	Turn	Medium	11.64547, 77.85129	30 KM/Hr
33	Turn	Medium	11.66653, 77.85622	30 KM/Hr
34	Turn	Medium	11.67328, 77.86076	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	Medium	11.67379, 77.86237	30 KM/Hr
36	Turn	High	11.68471, 77.86877	15 KM/Hr
37	Turn	High	11.68463, 77.87262	15 KM/Hr
38	Turn	High	11.69478, 77.87577	15 KM/Hr
39	Turn	High	11.69549, 77.87452	15 KM/Hr
40	Turn	High	11.76019, 77.89047	15 KM/Hr
41	Turn	High	11.75964, 77.89200	15 KM/Hr
42	Turn	Medium	11.83085, 77.94351	30 KM/Hr
43	Turn	Medium	11.84055, 77.94650	30 KM/Hr
44	Turn	Medium	11.90067, 77.98270	30 KM/Hr
45	Turn	Medium	11.91899, 78.01292	30 KM/Hr
46	Turn	Medium	11.91864, 78.02529	30 KM/Hr
47	Turn	High	11.93662, 78.05407	15 KM/Hr
48	Turn	Medium	11.95531, 78.04698	30 KM/Hr
49	Turn	High	11.95854, 78.04990	15 KM/Hr
50	Turn	High	11.96015, 78.04945	15 KM/Hr
51	Turn	Medium	11.96266, 78.05115	30 KM/Hr
52	Turn	High	11.96433, 78.05113	15 KM/Hr
53	Turn	High	11.96542, 78.04938	15 KM/Hr
54	Turn	High	11.96650, 78.04927	15 KM/Hr
55	Turn	Medium	12.12036, 78.13284	30 KM/Hr
56	Turn	High	12.29579, 78.07370	15 KM/Hr
57	Turn	High	12.29681, 78.07576	15 KM/Hr
58	Turn	High	12.31217, 78.07830	15 KM/Hr
59	Turn	High	12.31360, 78.07464	15 KM/Hr
60	Turn	High	12.35427, 78.08075	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
1	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
2	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
4	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
5	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
6	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
7	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
8	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
9	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
10	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
11	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
12	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
13	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
14	hospital	Thoppur, Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
15	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45708, 77.82867



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46547, 77.82486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47251, 77.83307



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47574, 77.83491



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47614, 77.83416



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48033, 77.83503



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48043, 77.83583



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.50430, 77.83818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50074, 77.84550



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49946, 77.84592



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49855, 77.86542



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50898, 77.86029



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52519, 77.85535



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55499, 77.85656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55605, 77.85768



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56378, 77.85967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57919, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58037, 77.85749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60225, 77.84735



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62707, 77.84192



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63085, 77.84072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64073, 77.84627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64130, 77.84816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64547, 77.85129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66653, 77.85622



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67328, 77.86076



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67379, 77.86237



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68471, 77.86877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69478, 77.87577



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69549, 77.87452



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83085, 77.94351



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84055, 77.94650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90067, 77.98270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91899, 78.01292



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91864, 78.02529



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93662, 78.05407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



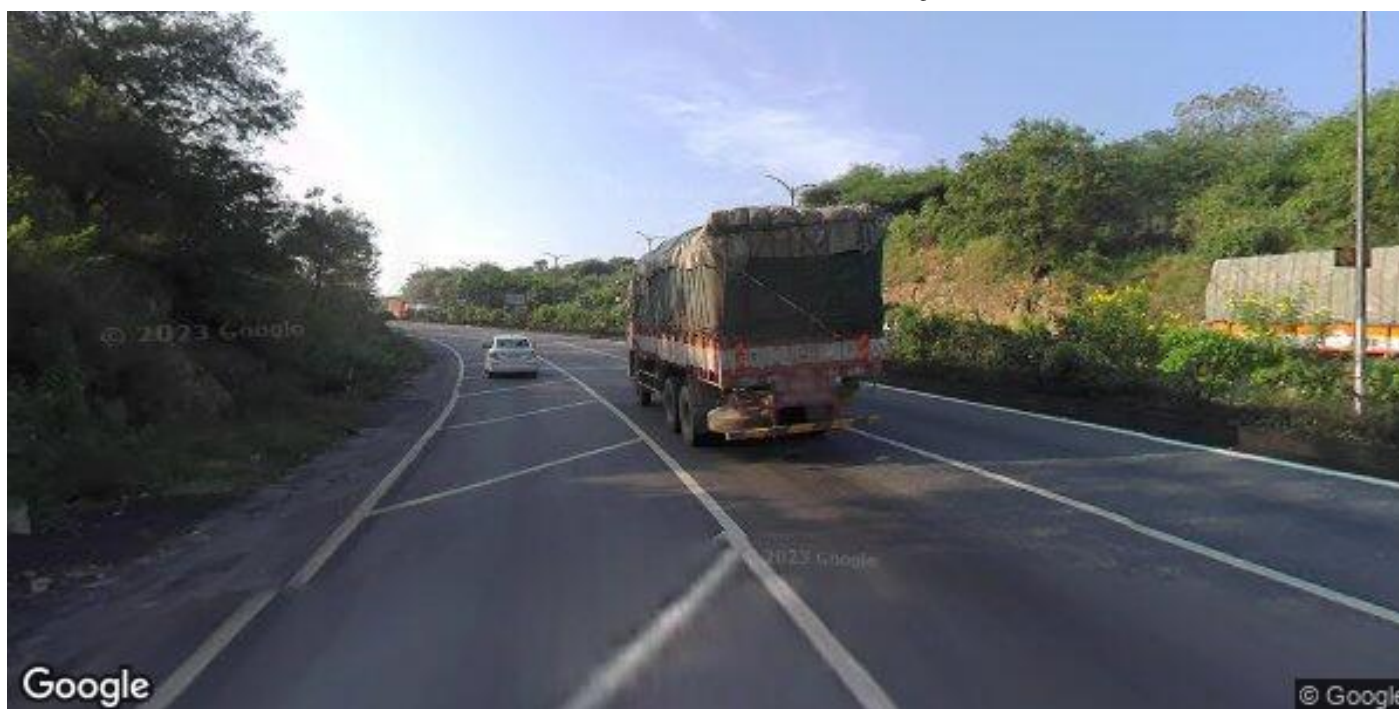
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12036, 78.13284



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29579, 78.07370



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29681, 78.07576



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31217, 78.07830



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31360, 78.07464



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35427, 78.08075

